1.

$$\int_0^1 x^2 + y^2 \ dx$$

2.

$$\int_0^1 \int_0^1 (x^2 + y^2) \, dx \, dy$$

3.

$$a_1^2 + a_2^2 = a_3^2$$

4.

$$x^{2\alpha} - 1 = y_{ij} + y_{ij}$$

5.

$$(a^n)^{r+s} = a^{nr+ns}$$

6.

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1 - p^{-s}}$$

7. Here are some examples of simple usage of subscripts and superscripts:

$$\int_0^1 x^2 + y^2 \ dx$$

Using superscript and subscripts in the same expression

$$a_1^2 + a_2^2 = a_3^2$$

Longer subscripts and superscripts:

$$x^{2\alpha} - 1 = y_{ij} + y_{ij}$$

Nested subscripts and superscripts

$$(a^n)^{r+s} = a^{nr+ns}$$

Example of a mathematical equation with subscripts and superscripts

$$\sum_{i=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1 - p^{-s}}$$

Squared root usage

$$\sqrt[4]{4ac} = \sqrt{4ac}\sqrt{4ac}$$